Therefore, using the in-degree centrality, we can conclude that the COLE framework will converge to the local optimal strategy at a Q-sublinear rate. □

Appendix C. Experimental Details of COLE_{SV}

This paper utilizes Proximal Policy Optimization (PPO) (Schulman et al., 2017) as the oracle algorithm for our set of strategies $\mathcal{N}$, which consists of convolutional neural network parameterized strategies. Each network is composed of 3 convolution layers with 25 filters and 3 fully-connected layers with 64 hidden neurons. To manage computational resources, we maintain a population size of 50 strategies. In instances where the population exceeds this limit, we randomly select one of the earliest ten removal strategies.

We run and evaluate all our experiments on Linux servers, which include two types of nodes: 1) 1-GPU node with NVIDIA GeForce 3090Ti 24G as GPU and AMD EPYC 7H12 64-Core Processor as CPU, 2) 2-GPUs node with GeForce RTX 3090 24G as GPU and AMD Ryzen Threadripper 3970X 32-Core Processor as CPU. On the Overcooked game environment, COLE_{SV} takes one to two days on the 2-GPUs machine for one layout's training.

The hyperparameter setup is similar to those in PBT and MEP, which are given as follows.

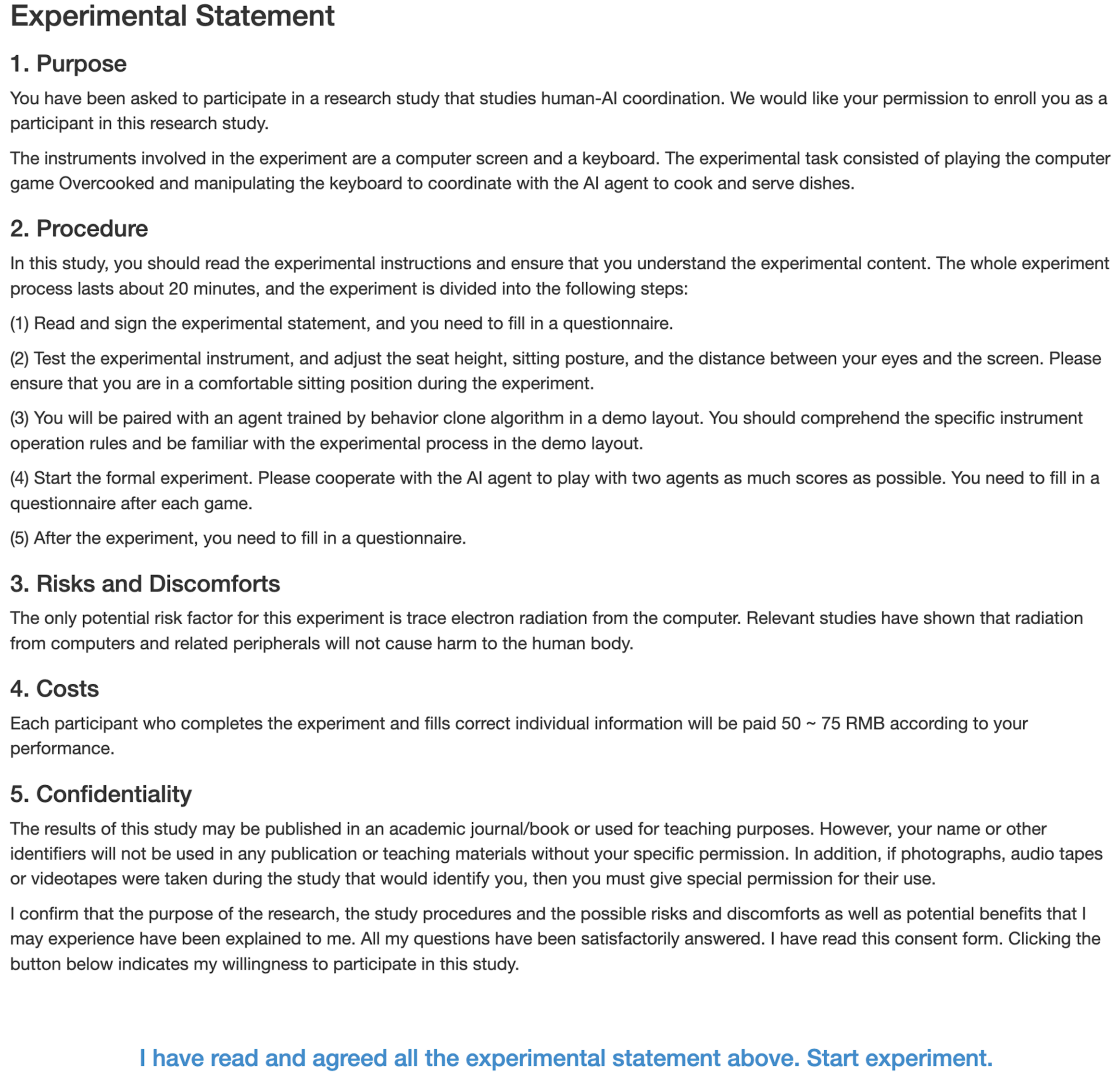
- The learning rate for each layout is 2e-3 , 1e-3 , 6e-4 , 8e-4 , and 8e-4.
- The gamma γ is 0.99.
- The lambda λ is 0.98.
- The PPO clipping factor is 0.05.
- The VF coefficient is 0.5.
- The maximum gradient norm is 0.1.
- The total training time steps for each PPO update is 48000, divided into 10 mini-batches.
- The total numbers of generations for each layout are 80, 60, 75, 70, and 70, respectively.
- For each generation, we update 10 times to approximate the best-preferred strategy.
- The α is 1.

Appendix D. Implementations of Baselines

In this part, we will introduce the detailed implementations of baselines. We train and evaluate self-play and PBT based on the Human-Aware Reinforcement Learning repository† (Carroll et al., 2020) and used Proximal Policy Optimization (PPO) (Schulman et al., 2017) as the RL algorithm. We implement FCP according to the FCP paper (Strouse et al., 2021) and use PPO as the RL algorithm. The implementation is based on the Human-Aware Reinforcement

†. `https://github.com/HumanCompatibleAI/human_aware_rl/tree/neurips2019`.

Learning repository (the same used in the self-paly and PBT). The MEP agent is trained with population size as 5, following the MEP paper (Zhao et al., 2021) and used the original implementation‡.



Experimental Statement

1. Purpose

You have been asked to participate in a research study that studies human-AI coordination. We would like your permission to enroll you as a participant in this research study.

The instruments involved in the experiment are a computer screen and a keyboard. The experimental task consisted of playing the computer game Overcooked and manipulating the keyboard to coordinate with the AI agent to cook and serve dishes.

2. Procedure

In this study, you should read the experimental instructions and ensure that you understand the experimental content. The whole experiment process lasts about 20 minutes, and the experiment is divided into the following steps:

(1) Read and sign the experimental statement, and you need to fill in a questionnaire.

(2) Test the experimental instrument, and adjust the seat height, sitting posture, and the distance between your eyes and the screen. Please ensure that you are in a comfortable sitting position during the experiment.

(3) You will be paired with an agent trained by behavior clone algorithm in a demo layout. You should comprehend the specific instrument operation rules and be familiar with the experimental process in the demo layout.

(4) Start the formal experiment. Please cooperate with the AI agent to play with two agents as much scores as possible. You need to fill in a questionnaire after each game.

(5) After the experiment, you need to fill in a questionnaire.

3. Risks and Discomforts

The only potential risk factor for this experiment is trace electron radiation from the computer. Relevant studies have shown that radiation from computers and related peripherals will not cause harm to the human body.

4. Costs

Each participant who completes the experiment and fills correct individual information will be paid 50 ~ 75 RMB according to your performance.

5. Confidentiality

The results of this study may be published in an academic journal/book or used for teaching purposes. However, your name or other identifiers will not be used in any publication or teaching materials without your specific permission. In addition, if photographs, audio tapes or videotapes were taken during the study that would identify you, then you must give special permission for their use.

I confirm that the purpose of the research, the study procedures and the possible risks and discomforts as well as potential benefits that I may experience have been explained to me. All my questions have been satisfactorily answered. I have read this consent form. Clicking the button below indicates my willingness to participate in this study.

I have read and agreed all the experimental statement above. Start experiment.

Figure 13: Screenshots of the Human-AI Experiment Platform - Experiment Statement.

Appendix E. Visual Overview of the Human-AI Experiment Platform

In this section, we provide a visual representation of the Human-AI Experiment Platform to offer readers a better understanding of the system's user interface, layout, and functionality. By including screenshots of various stages of the experiment process in Fig. 13, Fig. 14,

‡. The code of MEP original implementation: `https://github.com/ruizhaogit/maximum_entropy_population_based_training`.

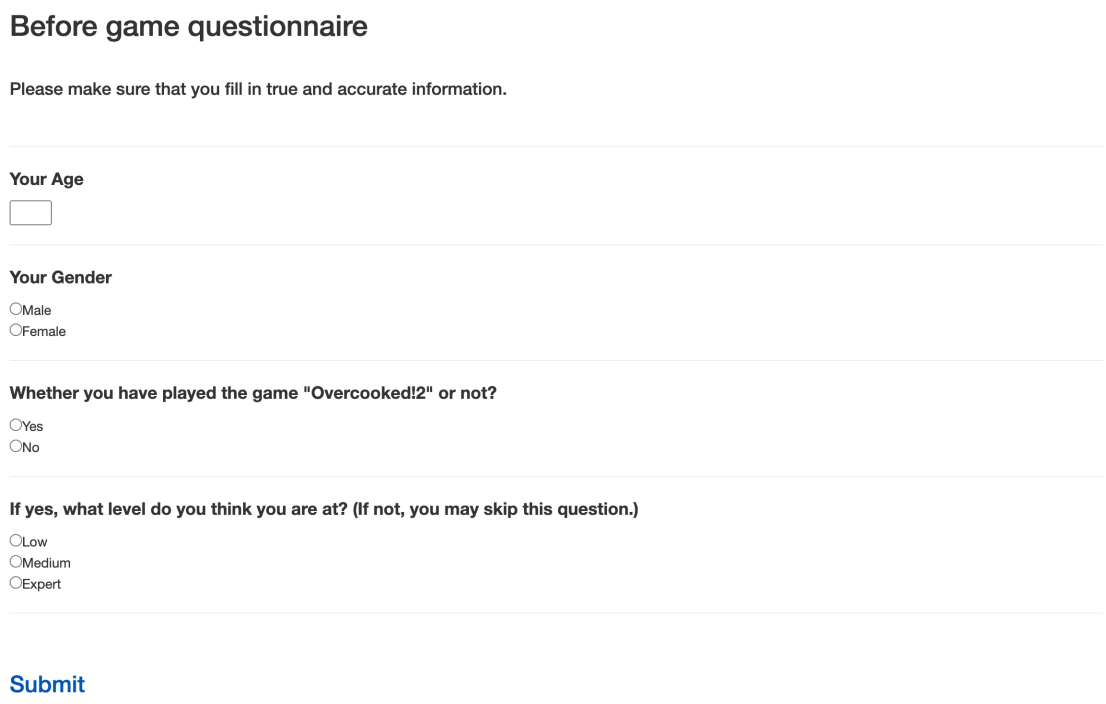


Figure 14: Screenshots of the Human-AI Experiment Platform - participant information questionnaire.

Fig. 15, Fig. 16, Fig. 17, Fig. 18, and Fig. 19, we aim to provide a comprehensive overview of the platform, enabling readers to better grasp its design and implementation.